

100 Days of Azure DevOps

Day 91 Handout — A public demo slice you can walk in five minutes — not a new ecommerce

10 - Portfolio & Public Launch · Personal lab only · Educational content · Not a sales pitch

Architecture A — public demo-shipboard (not employer work)

Box	Personal Azure object	What a stranger sees
Repo	Personal Azure Repos or GitHub: capstone/	README diagram + /health source
CI	Azure Pipeline on ubuntu-latest	Restore, test, publish zip/artifact
Artifact	PublishPipelineArtifact@1 named drop	Same bytes that will be promoted
Dev env	Azure App Service F1 or Container Apps consumption	https://<app>-dev.azurewebsites.net/health
Staging	Second slot or second app + Environment approval	Same artifact, different URL
Identity	Project-scoped service connection	No user PAT in YAML

Architecture B — scope control (In vs Out)

Idea	In (MVP)	Out (ambition)
App shape	One service: GET / and GET /health return build id + env	Payments, carts, recommendations ML
Topology	One repo folder capstone/	12 microservices and a service mesh
CI/CD	Tests + artifact + deploy + README diagram	Multi-region active-active
Data	In-memory or one small Azure Table/SQLite file	Cosmos + SQL + Redis 'because production'
Reuse	Harden an earlier 100-day app	New shop-clone repo on Day 91

Architecture C — capstone failure modes

Pitfall	What you see	Fix
Ambition spiral	Empty repo, grand README	Thin vertical slice: build→test→URL
Rebuild per environment	Dev and staging compile twice	Promote the same artifact / Build.BuildId
Employer app lift	Internal names, work screenshots	Public fictional demo only
No diagram	You cannot demo in five minutes	Four boxes: repo, pipeline, env, app
Secret in the repo	Connection string in appsettings	Key Vault / pipeline secret; F1 can still read KV refs

One-liner to remember

Capstones fail from ambition — ship a thin vertical slice you can demo in five minutes.

Step-by-step lab (personal Azure / Azure DevOps only)

Public demo system on a personal subscription and personal org. Work in capstone/. Do not start a giant ecommerce. Do not import employer code.

- 1 Create capstone/README.md with In/Out lists and a four-box diagram: repo → pipeline → environment → App Service (or Container Apps).
- 2 Reuse an earlier lab app or a tiny new demo-shipboard that returns { status, env, buildId } on GET /health. One service only.

- 3 Add or harden azure-pipelines.yml: trigger on capstone/**, ubuntu-latest, test, PublishPipelineArtifact@1.
- 4 Deploy that artifact to a personal F1 App Service or consumption Container App. Record the public /health URL.
- 5 Add a staging promotion path (slot or second app) that reuses the same artifact. Approval can be you.
- 6 Walk the demo in five minutes: PR or push → green run → URL shows the build id. If you cannot, cut scope.
- 7 No payments, no ML, no 12 services. If you want to add one, read the Out list out loud.

MVP pipeline — one service, tests, artifact, deploy

```
trigger:
  paths:
    include: [capstone/**]
pool:
  vmImage: ubuntu-latest
stages:
- stage: CI
  jobs:
  - job: build
    steps:
    - script: |
        test -f capstone/README.md
        echo CI for demo-shipboard $(Build.BuildId)
      displayName: Prove repo + build id
    - task: PublishPipelineArtifact@1
      inputs:
        targetPath: capstone
        artifact: drop
```

Done checklist

- capstone/ exists with In/Out and a diagram
- Public /health (or equivalent) on a personal Azure app
- CI publishes an artifact; staging reuses it
- No employer code, no ecommerce scaffold
- Five-minute demo path written in the README

Tomorrow — Day 92

Capstone 2 — the same public demo in a container tuxedo, not a Netflix clone